ASSIGNMENT-1

Foundations of Cryptography

Course: BCA | Subject: Cryptography

Q1. Define cryptography in your own words. What are the three main goals it tries to achieve?

Cryptography is the method of protecting information by transforming it into a form that unauthorized people cannot understand. Three important goals are confidentiality (only authorized people can read the data), integrity (data is not changed without detection), and authentication (confirming the identity of a user or the source of a message).

Q2. Differentiate between plaintext and ciphertext with a simple example.

Plaintext is the original readable message, while ciphertext is the scrambled form produced by encryption. For example, using a Caesar cipher with a shift of 3, plaintext “CAT” becomes ciphertext “FDW”.

Q3. Explain cryptography in detail and why it is essential in the modern digital world, giving at least three real-life applications.

Cryptography uses mathematical algorithms and keys to protect information during storage and communication. It helps keep private information confidential, detect unauthorized changes, verify identities, and support secure online transactions. It is essential because digital data travels through networks that may be observed or attacked. In online banking, cryptography protects login and transaction information. In messaging apps, end-to-end encryption helps prevent outsiders from reading conversations. In e-commerce, HTTPS protects payment and account details during checkout. It is also used in secure email, software updates, digital signatures, and virtual private networks.

Q4. What is the difference between encryption and decryption?

Encryption converts readable plaintext into ciphertext using an algorithm and key. Decryption reverses this process, using the appropriate key to recover the original plaintext from ciphertext.

Q5. Describe symmetric and asymmetric key cryptography. Compare key usage, speed, security and use cases, with examples.

Symmetric-key cryptography uses the same shared secret key for encryption and decryption. It is generally fast and suitable for large amounts of data, but the key must be shared securely. Examples include AES and DES (DES is now insecure for modern protection). It is used for file encryption, disk encryption and bulk data protection. Asymmetric-key cryptography uses a mathematically related public key and private key. The public key may be shared, while the private key must be protected. It is generally slower than symmetric encryption and is useful for key exchange, digital signatures and identity verification. Examples include RSA and ECC-based systems. In practice, secure protocols often combine both: asymmetric techniques establish or authenticate keys, and symmetric algorithms encrypt the actual data.

Q6. What is hashing, and how is it different from encryption?

Hashing transforms input data into a fixed-length digest using a hash function, such as SHA-256. Hashing is designed to be one-way and is used to check integrity; encryption is reversible with the correct key and is used to keep information confidential.

Q7. Critically evaluate why achieving confidentiality, integrity and non-repudiation together is challenging in real-world communication.

These goals require different mechanisms and careful key management. Encryption can provide confidentiality, while message authentication codes or authenticated encryption can protect integrity and authenticity. Non-repudiation usually requires digital signatures and reliable identity and certificate management. Compromised keys, poor implementation, stolen credentials, and disputes about who controlled a key can weaken these guarantees, so no single technique automatically provides all three in every situation.

Q8. Explain the role of cryptography in network security: confidentiality, integrity, authentication and non-repudiation, with an example for each.

Cryptography protects data as it travels across networks. Confidentiality: TLS encrypts web traffic so an eavesdropper cannot easily read it. Integrity: a message authentication code or authenticated encryption detects unauthorized changes to transmitted data. Authentication: TLS certificates help a browser verify that it is communicating with the intended website. Non-repudiation: a digital signature on a document can provide evidence that the holder of a private key signed it, subject to trustworthy identity verification and key control. These protections depend on correct configuration, secure keys

and reliable implementations.

Q9. Name two network security protocols that use cryptographic techniques and state their purpose.

1. SSL/TLS: TLS protects network communication by supporting authentication, confidentiality and integrity; SSL is obsolete and should not be used. 2. IPsec: IPsec protects IP traffic, commonly by encrypting and/or authenticating packets in VPN connections.

Q10. Critically discuss the evolution of cryptography from classical ciphers to modern systems, considering computational advances.

Classical cryptography used substitution and transposition methods, such as Caesar and Vigenère ciphers. These could often be broken through language patterns and frequency analysis. Mechanical systems such as Enigma made encryption more complex, but wartime cryptanalysis and computing demonstrated the importance of mathematical analysis and operational security. Modern cryptography uses carefully studied algorithms, larger keys and computer-based protocols. DES became a major standard but its 56-bit key became vulnerable to brute-force search; AES replaced it with stronger key options and a larger block size. Public-key techniques such as RSA enabled key establishment and digital signatures without requiring every pair of users to share a secret in advance. Faster computers help defenders implement strong cryptography, but also help attackers test keys and exploit weak algorithms. Therefore, systems must use current standards, adequate key sizes, secure implementations and timely updates.

Q11. Trace the historical development from Caesar and Vigenère through Enigma, public-key cryptography, DES, AES and RSA.

Early methods included the Caesar cipher, which shifts letters, and the Vigenère cipher, which applies shifts based on a keyword. Both have weaknesses that can be exploited by statistical analysis. During World War II, the Enigma rotor machine was used to protect military communications; its defeat involved mathematical work, computing, captured information and procedural weaknesses. In the 1970s, public-key cryptography emerged, allowing public and private keys to support key exchange and digital signatures; Diffie and Hellman published their key-exchange work in 1976, and RSA was introduced in 1977. DES was adopted as a U.S. encryption standard in 1977, but its short effective key later became inadequate. The AES competition selected Rijndael, which became the AES standard in 2001. Modern systems commonly use AES for data encryption and public-key or elliptic-curve techniques for key establishment and signatures.

Q12. Who introduced the concept of public-key cryptography, and in which year?

Whitfield Diffie and Martin Hellman publicly introduced the idea of public-key cryptography in 1976. The independently developed RSA public-key algorithm was published by Rivest, Shamir and Adleman in 1977.

References

1. William Stallings, Cryptography and Network Security: Principles and Practice.

2. NIST publications on DES, AES and cryptographic standards.

3. Whitfield Diffie and Martin E. Hellman, New Directions in Cryptography, 1976.